

- 3 (a) (i) (vertical component = $44 \sin 30^\circ =$) 22 N A1 [1]
- (ii) (horizontal component = $44 \cos 30^\circ =$) 38(.1) N A1 [1]
- (b) $W \times 0.64 = 22 \times 1.60$ C1
- (W =) 55 N A1 [2]
- (c) F has a horizontal component (not balanced by W)
 or F has 38 N acting horizontally
 or 38 N acts on wall
 or vertical component of F does not balance W
 or F and W do not make a closed triangle of forces B1 [1]
- (d) line from P in direction towards point on wire vertically above W and direction up B1 [1]